(i) the consumer's surplus is 27 units

(ii) the producer's surplus is 9 units.

Exercise 3.3

1. Calculate consumer's surplus if the demand function $p = 50 - 2x$ and $x = 20$
2. Calculate consumer's surplus if the demand function $p = 122 - 5x - 2x^2$ and $x = 6$
3. The demand function $p = 85 - 5x$ and supply function $p = 3x - 35$. Calculate the equilibrium price and quantity demanded .Also calculate consumer's surplus.
4. The demand function for a commodity is $p = e^{-x}$. Find the consumer's surplus when $p = 0.5$.
5. Calculate the producer's surplus at $x = 5$ for the supply function $p = 7 + x$.
6. If the supply function for a product is $p = 3x + 5x^2$.Find the producer's surplus when $x = 4$.
7. The demand function for a commodity is $p = \frac{36}{x+4}$. Find the consumer's surplus when the prevailing market price is ₹6.
8. The demand and supply functions under perfect competition are $p_d = 1600 - x^2$ and $p_s = 2x^2 + 400$ respectively. Find the producer's surplus.
9. Under perfect competition for a commodity the demand and supply laws are $p_d = \frac{8}{x+1} - 2$ and $p_s = \frac{x+3}{2}$ respectively. Find the consumer's and producer's surplus.
10. The demand equation for a products is $x = \sqrt{100 - p}$ and the supply equation is $x = \frac{p}{2} - 10$. Determine the consumer's surplus and producer's surplus, under market equilibrium.
11. Find the consumer's surplus and producer's surplus for the demand function $p_d = 25 - 3x$ and supply function $p_s = 5 + 2x$.

Exercise 3.4

Choose the best answer form the given alternatives

1. Area bounded by the curve $y = x(4 - x)$ between the limits 0 and 4 with $x-$ axis is

   (a) $\frac{30}{3}$ sq.units (b) $\frac{31}{2}$ sq.units

   (c) $\frac{32}{3}$ sq.units (d) $\frac{15}{2}$ sq.units

2. Area bounded by the curve $y = e^{-2x}$ between the limits $0 \le x \le \infty$ is

   (a)1 sq.units (b) $\frac{1}{2}$ sq.unit

   (c) 5 sq.units (d) 2 sq.units

3. Area bounded by the curve $y = \frac{1}{x}$ between the limits 1 and 2 is

   (a) log2 sq.units (b) log5 sq.units

   (c) log3 sq.units (d) log 4 sq.units

4. If the marginal revenue function of a firm is MR= $e^{\frac{-x}{10}}$, then revenue is

   (a) $-10e^{\frac{-x}{10}}$ (b) $1 - e^{\frac{-x}{10}}$

   (c) $10\left(1 - e^{\frac{-x}{10}}\right)$ (d) $e^{\frac{-x}{10}} + 10$

   J3FXVR

5. If MR and MC denotes the marginal revenue and marginal cost functions, then the profit functions is

   (a) $P = \int (MR - MC)\, dx + k$

   (b) $P = \int (MR + MC)\, dx + k$

   (c) $P = \int (MR)(MC) dx + k$

   (d) $P = \int (R - C) dx + k$

XII Std - Business Maths & Stat EM Chapter 3.indd 75 2/28/2020 5:49:52 PM